

Prismatic: The Adaptive Cognitive Engine

Refracting Complex Knowledge into 4 Personalized Learning Modalities on Amazon Quick

FOUNDING TEAM & AFFILIATION:

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THE 2:00 AM COGNITIVE PARADOX:

It's 2:00 AM before a midterm. A student re-reads the same paragraph six times, asks ChatGPT, and gets an 800-word text wall that heightens panic. The issue isn't intelligence—it's wavelength. Lectures broadcast in monochrome, while human brains learn in spectrums.



SCAN FOR DEMO

GitHub repo, web app code,
prompt specs & cue cards.

< 30 SECONDS

Cognitive Triage

5-question diagnostic maps intake
profile

4 SPECTRUMS

Learning Modalities

Analogies, Schemas, Socratic &
Decks

2-SENTENCE CAP

Socratic Sparring

Active recall without text dumps

100% NO-CODE

Amazon Quick

Full multi-widget pipeline in 60
mins

The 4 Learning Spectrums: Zero Errors, Pure Retention

Orchestrated in a 60-minute build sprint on Amazon Quick & Amazon Bedrock foundation models

METAPHORICAL

Dorm & Dining Hall Realities

SPECTRUM 1: INTUITIVE ANALOGY

Translates abstract theory into daily campus navigation. E.g., Dijkstra's shortest path = navigating peak dining hall line rush at Blackburn Center.

VISUO-SPATIAL

Structural Concept Maps & ASCII

SPECTRUM 2: VISUAL SCHEMATIC

Renders execution trees, logical branches, and state machines into structural mental schemas for complex STEM and CS coursework.

KINESTHETIC

Active Retrieval Partner

SPECTRUM 3: SOCRATIC SPARRING

Chatbot strictly capped at 2 sentences; never lectures; fires progressive edge cases to force active recall without giving away answers.

SYNTHESIS

4-Slide Micro-Presentation

SPECTRUM 4: EXAM TRAP DECK

Distills the topic into 4 high-yield slides exposing the exact trick questions professors test on Midterm #2, paired with AI visual anchors.

Why Prismatic Wins: The \$80/Hour Tutoring Alternative

Comparing modern AI study workflows against passive text dumps and expensive human tutoring

VS. CHATGPT & RAW LLMS

Active Sparring vs. Text Dumps

ChatGPT dumps 800-word walls students glaze over. Prismatic caps replies at 2 sentences and forces active recall.

VS. \$80/HR PRIVATE TUTORING

Democratizing Academic Mastery

Tutors cost \$80/hr; office hours are 2 hrs/week. Prismatic gives every student 24/7 adaptive tutoring for pennies.

VS. STATIC SLIDES & SYLLABI

Dynamic Cognitive Fit

Ends task paralysis for ADHD, visual, and verbal learners with a 30-second diagnostic.

'With More Time We'd Add...' & The Shark Tank Verdict

Meeting 100% of official judging criteria: Creativity, Usefulness, Execution & Teamwork

CREATIVITY

Cognitive refraction metaphor
(white light → spectrums)

USEFULNESS

Directly solves dorm paralysis &
exam anxiety

EXECUTION

Working 6-widget pipeline on
Amazon Quick

TEAMWORK

10 teammates with dedicated active
roles

OFFICIAL HANDOUT MANDATE: 'WITH MORE TIME WE'D ADD...'

"With more time, we'd add live Canvas and syllabus LMS integration, automatically syncing weekly exam dates and generating proactive 15-minute micro-sprints before every morning class.

We don't just help students pass exams; we eliminate academic paralysis and unlock confidence."

WE ARE PRISMATIC — REFRACTING COMPLEX KNOWLEDGE INTO YOUR NATIVE SPECTRUM. VOTE PRISMATIC!